

Natural Logarithm Integral Def

The natural logarithm is usually introduced as the inverse of the exponential function e^x , but there is a cleaner, self-contained way to *define* it that reveals exactly where it comes from and why it behaves the way it does.

Definition. For any $x > 0$,

$$\ln x = \int_1^x \frac{1}{t} dt$$

Geometrically, $\ln x$ is the signed area under the curve $y = 1/t$ from $t = 1$ to $t = x$. When $x > 1$ the area is positive; when $0 < x < 1$ the area is negative (you are integrating leftward from 1); and $\ln 1 = 0$ because the interval has zero width.

This definition immediately explains the derivative: by the Fundamental Theorem of Calculus, if $f(x) = \int_1^x \frac{1}{t} dt$, then $f'(x) = \frac{1}{x}$. So $\frac{d}{dx} \ln x = \frac{1}{x}$ —no circular reasoning, no hand-waving.

Worked example. Show that $\ln(ab) = \ln a + \ln b$ for $a, b > 0$.

Split the integral at $t = a$:

$$\ln(ab) = \int_1^{ab} \frac{dt}{t} = \int_1^a \frac{dt}{t} + \int_a^{ab} \frac{dt}{t}$$

In the second integral, substitute $t = au$ so $dt = a du$. When $t = a$, $u = 1$; when $t = ab$, $u = b$:

$$\int_a^{ab} \frac{dt}{t} = \int_1^b \frac{a du}{au} = \int_1^b \frac{du}{u} = \ln b$$

Therefore $\ln(ab) = \ln a + \ln b$. The product rule for logarithms is a direct consequence of how the integral scales under substitution—it is built into the definition, not a separate fact to memorize.

Problem-solving application. Evaluate $\int_1^{e^3} \frac{2}{x} dx$.

Recognize $\frac{1}{x}$ as the integrand of $\ln x$:

$$\int_1^{e^3} \frac{2}{x} dx = 2 \ln x \Big|_1^{e^3} = 2 \ln(e^3) - 2 \ln 1 = 2(3) - 0 = 6$$

More broadly, whenever you see $\int \frac{f'(x)}{f(x)} dx$, the substitution $u = f(x)$ converts it to $\int \frac{du}{u} = \ln |u| + C$. Recognizing this pattern—a derivative in the numerator over the original function—is one of the most frequently tested integration techniques in calculus.